

The empirical recurrence for OEIS A295603: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A295603 records “Empirical recurrence of order 69”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 87$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295603 is “Number of $n \times 5$ 0..1 arrays with each 1 horizontally or vertically adjacent to 0, 2, 3 or 4 1s.”. It has offset 1 and begins

13, 132, 1425, 16340, 201602, 2594128, 34416318, 465449155, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Nov 24 2017 07:22:56, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 87$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 87$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 174$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	25	c_{19}	−7886921638	c_{37}	81970633315717	c_{55}	−8734403068
c_2	−189	c_{20}	−36841106669	c_{38}	−7226792089710	c_{56}	46434758513
c_3	552	c_{21}	2422562862	c_{39}	−75135968920622	c_{57}	−56196157359
c_4	−828	c_{22}	82688421593	c_{40}	−113189552856111	c_{58}	−45203338640
c_5	−2840	c_{23}	−83999035915	c_{41}	−71941530219386	c_{59}	5229800236
c_6	21532	c_{24}	−476389636416	c_{42}	57149495139141	c_{60}	380764820
c_7	−153905	c_{25}	292999377630	c_{43}	114382261699368	c_{61}	−149269586
c_8	577671	c_{26}	2815301304752	c_{44}	12351407047260	c_{62}	−443541094
c_9	−351666	c_{27}	1331559553407	c_{45}	−78387333440373	c_{63}	71870881
c_{10}	964915	c_{28}	−8413193820408	c_{46}	−23580161756243	c_{64}	196075026
c_{11}	14584019	c_{29}	−8409203834148	c_{47}	36517103469373	c_{65}	−56029578
c_{12}	−17623398	c_{30}	19398667130664	c_{48}	24504040945996	c_{66}	−27458105
c_{13}	135032297	c_{31}	36785891003893	c_{49}	−6450106132995	c_{67}	5867106
c_{14}	74467308	c_{32}	−15493739630398	c_{50}	−11805304904918	c_{68}	867728
c_{15}	−30681619	c_{33}	−90959573280138	c_{51}	−1372295231716	c_{69}	−147286
c_{16}	736518965	c_{34}	−61316169651664	c_{52}	1629978307992		
c_{17}	−783634971	c_{35}	60201903129158	c_{53}	288610677492		
c_{18}	1594318543	c_{36}	125759226479263	c_{54}	108152476152		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

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